1) a)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}-3\\8\end{pmatrix} - \begin{pmatrix}2\\6\end{pmatrix} = \begin{pmatrix}-5\\2\end{pmatrix}$

1) b)

$\left|\overrightarrow{AB}\right| = \sqrt{(-5)^2 + 2^2} = \sqrt{29}$

1) c)

Let $\hat{\mathbf{p}}$ be the unit vector in the direction of $\overrightarrow{AB}$

$\hat{\mathbf{p}} = \frac{\overrightarrow{AB}}{\left|\overrightarrow{AB}\right|} = \frac{1}{\sqrt{29}}\begin{pmatrix}-5\\2\end{pmatrix} = \begin{pmatrix}-\frac{5}{\sqrt{29}}\\ \frac{2}{\sqrt{29}}\end{pmatrix} = \begin{pmatrix}-\frac{5\sqrt{29}}{29}\\ \frac{2\sqrt{29}}{29}\end{pmatrix}$

2)

$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = (7\mathbf{i} + 8\mathbf{j}) - (4\mathbf{i} - \mathbf{j}) = 3\mathbf{i} + 9\mathbf{j}$

$\left|\overrightarrow{PQ}\right| = \sqrt{3^2 + 9^2} = \sqrt{90} = 3\sqrt{10}$

Let $\hat{\mathbf{a}}$ be the unit vector in the direction of $\overrightarrow{PQ}$

$\hat{\mathbf{a}} = \frac{\overrightarrow{PQ}}{\left|\overrightarrow{PQ}\right|} = \frac{1}{3\sqrt{10}}(3\mathbf{i} + 9\mathbf{j})$

Then a vector of length $6\sqrt{10}$ in the direction of $\overrightarrow{PQ}$ is

$6\sqrt{10} \times \frac{1}{3\sqrt{10}}(3\mathbf{i} + 9\mathbf{j}) = 2(3\mathbf{i} + 9\mathbf{j}) = 6\mathbf{i} + 18\mathbf{j}$

3)

$\overrightarrow{ST} = \overrightarrow{OT} - \overrightarrow{OS} = \begin{pmatrix}\sqrt{96}\\\sqrt{98}\end{pmatrix} - \begin{pmatrix}\sqrt{150}\\\sqrt{162}\end{pmatrix} = \begin{pmatrix}4\sqrt{6}\\7\sqrt{2}\end{pmatrix} - \begin{pmatrix}5\sqrt{6}\\9\sqrt{2}\end{pmatrix} = \begin{pmatrix}-\sqrt{6}\\-2\sqrt{2}\end{pmatrix}$

$\left|\overrightarrow{ST}\right| = \sqrt{\left(-\sqrt{6}\right)^2 + \left(-2\sqrt{2}\right)^2} = \sqrt{14}$

Let $\hat{\mathbf{a}}$ be the unit vector in the direction of $\overrightarrow{ST}$

$\hat{\mathbf{a}} = \frac{\overrightarrow{ST}}{\left|\overrightarrow{ST}\right|} = \frac{1}{\sqrt{14}}\begin{pmatrix}-\sqrt{6}\\-2\sqrt{2}\end{pmatrix}$

Then a vector of length $\sqrt{126} = 3\sqrt{14}$ in the direction of $\overrightarrow{ST}$ is

$3\sqrt{14} \times \frac{1}{\sqrt{14}}\begin{pmatrix}-\sqrt{6}\\-2\sqrt{2}\end{pmatrix} = 3\begin{pmatrix}-\sqrt{6}\\-2\sqrt{2}\end{pmatrix} = \begin{pmatrix}-3\sqrt{6}\\-6\sqrt{2}\end{pmatrix}$

4) a)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \big((2k-3)\mathbf{i} + (3k+1)\mathbf{j}\big) - \big((k-6)\mathbf{i} + (2k+4)\mathbf{j}\big)$

$= (k+3)\mathbf{i} + (k-3)\mathbf{j}$

$\left|\overrightarrow{AB}\right| = \sqrt{(k+3)^2 + (k-3)^2} = \sqrt{k^2+6k+9+k^2-6k+9} = \sqrt{2k^2+18}$

Let $\hat{\mathbf{p}}$ be the unit vector in the direction of $\overrightarrow{AB}$

$\hat{\mathbf{p}} = \frac{\overrightarrow{AB}}{\left|\overrightarrow{AB}\right|} = \frac{1}{\sqrt{2k^2+18}}\big((k+3)\mathbf{i} + (k-3)\mathbf{j}\big)$

Then a vector of length $\sqrt{4k^2+36}$ in the direction of $\overrightarrow{AB}$ is

$\mathbf{p} = \sqrt{4k^2+36} \times \frac{1}{\sqrt{2k^2+18}}\big((k+3)\mathbf{i} + (k-3)\mathbf{j}\big)$

$= \frac{\sqrt{4k^2+36}}{\sqrt{2k^2+18}}\big((k+3)\mathbf{i} + (k-3)\mathbf{j}\big)$

$= \sqrt{\frac{4k^2+36}{2k^2+18}}\big((k+3)\mathbf{i} + (k-3)\mathbf{j}\big)$

$= \sqrt{\frac{4(k^2+9)}{2(k^2+9)}}\big((k+3)\mathbf{i} + (k-3)\mathbf{j}\big)$

$= \sqrt{2}\big((k+3)\mathbf{i} + (k-3)\mathbf{j}\big)$

$\therefore \mathbf{p} = \big(\sqrt{2}k + 3\sqrt{2}\big)\mathbf{i} + \big(\sqrt{2}k - 3\sqrt{2}\big)\mathbf{j}$

4) b)

$|\mathbf{p}| = \sqrt{\big(\sqrt{2}k + 3\sqrt{2}\big)^2 + \big(\sqrt{2}k - 3\sqrt{2}\big)^2}$

$= \sqrt{(2k^2+12k+18) + (2k^2-12k+18)}$

$= \sqrt{4k^2+36}$

If k is a positive integer, then the smallest $|\mathbf{p}|$ can be is when $k = 1$

$\therefore \mathbf{p} = 4\sqrt{2}\mathbf{i} - 2\sqrt{2}\mathbf{j}$

5) a)

$\mathbf{a} \cdot \mathbf{b} = \begin{pmatrix}2\\3\end{pmatrix} \cdot \begin{pmatrix}3\\5\end{pmatrix} = 2 \times 3 + 3 \times 5 = 6 + 15 = 21$

5) b)

$\mathbf{a} \cdot \mathbf{b} = (4\mathbf{i} + \mathbf{j}) \cdot (-2\mathbf{i} + 5\mathbf{j}) = 4 \times (-2) + 1 \times 5 = -8 + 5 = -3$

5) c)

$\mathbf{a} \cdot \mathbf{b} = \begin{pmatrix} 0 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ -6 \end{pmatrix} = 0 \times (-3) + (-4) \times (-6) = 0 + 24 = 24$

5) d)

$\mathbf{a} \cdot \mathbf{b} = (0\mathbf{i} + 9\mathbf{j}) \cdot (6\mathbf{i} - \mathbf{j}) = 0 \times 6 + 9 \times (-1) = 0 - 9 = -9$

6)

$(3p\mathbf{i} - 2\mathbf{j}) \cdot (-5\mathbf{i} - 4p\mathbf{j}) = (3p) \times (-5) + (-2) \times (-4p) = -15p + 8p = -7p$

7)

$\left(8q^3\mathbf{i} + \frac{5}{q}\mathbf{j}\right) \cdot \left(-\frac{3}{q^2}\mathbf{i} + 9q^2\mathbf{j}\right) = (8q^3) \times \left(-\frac{3}{q^2}\right) + \left(\frac{5}{q}\right) \times (9q^2) = -24q + 45q = 21q$

8) a)

$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{1\times4+9\times1}{\sqrt{1^2+9^2}\times\sqrt{4^2+1^2}} = \frac{13}{\sqrt{82}\sqrt{17}} = \frac{13\sqrt{1394}}{1394}$

$\theta = \arccos\left(\frac{13\sqrt{1394}}{1394}\right) = 69.6°$ (1 DP)

8) b)

$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{(-2)\times3+10\times9}{\sqrt{(-2)^2+10^2}\times\sqrt{3^2+9^2}} = \frac{84}{\sqrt{104}\sqrt{90}} = \frac{7\sqrt{65}}{65}$

$\theta = \arccos\left(\frac{7\sqrt{65}}{65}\right) = 29.7°$ (1 DP)

8) c)

$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{(-2)\times10+(-11)\times3}{\sqrt{(-2)^2+(-11)^2}\times\sqrt{10^2+3^2}} = \frac{-53}{\sqrt{125}\sqrt{109}} = -\frac{53\sqrt{545}}{2725}$

$\theta = \arccos\left(-\frac{53\sqrt{545}}{2725}\right) = 117.0°$ (1 DP)

8) d)

$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{(-8)\times(-10)+1\times2}{\sqrt{(-8)^2+1^2}\times\sqrt{(-10)^2+2^2}} = \frac{82}{\sqrt{65}\sqrt{104}} = \frac{41\sqrt{10}}{130}$

$\theta = \arccos\left(\frac{41\sqrt{10}}{130}\right) = 4.2°$ (1 DP)

9)

$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \begin{pmatrix}2\\4\end{pmatrix} - \begin{pmatrix}-2\\1\end{pmatrix} = \begin{pmatrix}4\\3\end{pmatrix}$

$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{pmatrix}-1\\5\end{pmatrix} - \begin{pmatrix}-2\\1\end{pmatrix} = \begin{pmatrix}1\\4\end{pmatrix}$

$\cos\theta = \frac{\overrightarrow{BA}\cdot\overrightarrow{BC}}{|\overrightarrow{BA}||\overrightarrow{BC}|} = \frac{4\times1+3\times4}{\sqrt{4^2+3^2}\times\sqrt{1^2+4^2}} = \frac{16}{\sqrt{25}\sqrt{17}} = \frac{16\sqrt{17}}{85}$

$\theta = \arccos\left(\frac{16\sqrt{17}}{85}\right) = 39.1°$ (1 DP)

10)

If $\mathbf{a}$ and $\mathbf{b}$ are perpendicular, then $\mathbf{a}\cdot\mathbf{b} = 0$

11) a)

$\mathbf{a}\cdot\mathbf{b} = 3\times(-1) + 1\times(-3) = -3 - 3 = -6$

$\therefore$ $\mathbf{a}$ and $\mathbf{b}$ are not perpendicular

11) b)

$\mathbf{a}\cdot\mathbf{b} = 2\times(-2) + (-5)\times 5 = -4 - 25 = -29$

$\therefore$ $\mathbf{a}$ and $\mathbf{b}$ are not perpendicular

11) c)

$\mathbf{a}\cdot\mathbf{b} = (-8)\times 2 + 2\times 8 = -16 + 16 = 0$

$\therefore$ $\mathbf{a}$ and $\mathbf{b}$ are perpendicular

11) d)

$\mathbf{a}\cdot\mathbf{b} = \sqrt{8}\times\sqrt{72} + \left(-\sqrt{32}\right)\times\sqrt{18} = 2\sqrt{2}\times 6\sqrt{2} - 4\sqrt{2}\times 3\sqrt{2} = 24 - 24 = 0$

$\therefore$ $\mathbf{a}$ and $\mathbf{b}$ are perpendicular

12) a)

$\mathbf{a}\cdot\mathbf{b} = 0 \Rightarrow (2p)\times 5 + 4\times(-3) = 0$

$\Rightarrow 10p - 12 = 0 \Rightarrow p = \frac{6}{5}$

12) b)

$\mathbf{a}\cdot\mathbf{b} = 0 \Rightarrow 6\times(8p) + (-4p)\times\frac{1}{2} = 0$

$\Rightarrow 48p - 2p = 0 \Rightarrow 46p = 0 \Rightarrow p = 0$

12) c)

$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow (4+p) \times (2p) + (3p) \times (1-p) = 0$
$\Rightarrow 8p + 2p^2 + 3p - 3p^2 = 0$
$\Rightarrow 11p - p^2 = 0 \Rightarrow (11-p)p = 0 \Rightarrow p = 0$ or $p = 11$

12) d)

$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow (2p-1) \times (p-2) + (3p-1) \times (2p+3) = 0$
$\Rightarrow 2p^2 - 5p + 2 + 6p^2 + 7p - 3 = 0$
$\Rightarrow 8p^2 + 2p - 1 = 0 \Rightarrow (4p-1)(2p+1) = 0 \Rightarrow p = \frac{1}{4}$ or $p = -\frac{1}{2}$

13)

$\mathbf{a} \cdot \mathbf{b} = (3k+2) \times (-6k-6) + (2k+2) \times (9k+6)$
$= -18k^2 - 30k - 12 + 18k^2 + 30k + 12 = 0$
$\therefore$ $\mathbf{a}$ and $\mathbf{b}$ are perpendicular for all values of k

14)

$\mathbf{a} \cdot \mathbf{b} = (k+1) \times (2-k) + (6k) \times (4k+1)$
$= 2k - k^2 + 2 - k + 24k^2 + 6k = 23k^2 + 7k + 2$
If $\mathbf{a}$ and $\mathbf{b}$ are perpendicular, then $\mathbf{a} \cdot \mathbf{b} = 0$
Consider $23k^2 + 7k + 2 = 0$
$b^2 - 4ac = 7^2 - 4 \times 23 \times 2 = -135 < 0$
As $23k^2 + 7k + 2 = 0$ has no real roots, $\mathbf{a}$ and $\mathbf{b}$ can never be perpendicular

15) a)

$\mathbf{r} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

15) b)

$\mathbf{r} = \begin{pmatrix} -2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 9 \end{pmatrix}$

15) c)

$\mathbf{r} = \begin{pmatrix} 0 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

15) d)

$\mathbf{r} = \begin{pmatrix} a \\ b \end{pmatrix} + \lambda \begin{pmatrix} p \\ q \end{pmatrix}$

16) a)

$\begin{pmatrix}1\\4\end{pmatrix}-\begin{pmatrix}2\\9\end{pmatrix}=\begin{pmatrix}-1\\-5\end{pmatrix}$

$\mathbf{r}=\begin{pmatrix}1\\4\end{pmatrix}+\lambda\begin{pmatrix}-1\\-5\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}2\\9\end{pmatrix}+\lambda\begin{pmatrix}-1\\-5\end{pmatrix}$

or $\mathbf{r}=\begin{pmatrix}1\\4\end{pmatrix}+\lambda\begin{pmatrix}1\\5\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}2\\9\end{pmatrix}+\lambda\begin{pmatrix}1\\5\end{pmatrix}$

16) b)

$\begin{pmatrix}3\\5\end{pmatrix}-\begin{pmatrix}5\\15\end{pmatrix}=\begin{pmatrix}-2\\-10\end{pmatrix}$

$\mathbf{r}=\begin{pmatrix}3\\5\end{pmatrix}+\lambda\begin{pmatrix}-2\\-10\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}5\\15\end{pmatrix}+\lambda\begin{pmatrix}-2\\-10\end{pmatrix}$

or $\mathbf{r}=\begin{pmatrix}3\\5\end{pmatrix}+\lambda\begin{pmatrix}1\\5\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}5\\15\end{pmatrix}+\lambda\begin{pmatrix}1\\5\end{pmatrix}$

16) c)

$\begin{pmatrix}-2\\1\end{pmatrix}-\begin{pmatrix}0\\-5\end{pmatrix}=\begin{pmatrix}-2\\6\end{pmatrix}$

$\mathbf{r}=\begin{pmatrix}-2\\1\end{pmatrix}+\lambda\begin{pmatrix}-2\\6\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}0\\-5\end{pmatrix}+\lambda\begin{pmatrix}-2\\6\end{pmatrix}$

or $\mathbf{r}=\begin{pmatrix}-2\\1\end{pmatrix}+\lambda\begin{pmatrix}-1\\3\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}0\\-5\end{pmatrix}+\lambda\begin{pmatrix}-1\\3\end{pmatrix}$

16) d)

$\begin{pmatrix}-4\\3\end{pmatrix}-\begin{pmatrix}4\\-13\end{pmatrix}=\begin{pmatrix}-8\\16\end{pmatrix}$

$\mathbf{r}=\begin{pmatrix}-4\\3\end{pmatrix}+\lambda\begin{pmatrix}-8\\16\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}4\\-13\end{pmatrix}+\lambda\begin{pmatrix}-8\\16\end{pmatrix}$

or $\mathbf{r}=\begin{pmatrix}-4\\3\end{pmatrix}+\lambda\begin{pmatrix}-1\\2\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}4\\-13\end{pmatrix}+\lambda\begin{pmatrix}-1\\2\end{pmatrix}$

16) e)

$\begin{pmatrix}2\\9\end{pmatrix}-\begin{pmatrix}-3\\-20\end{pmatrix}=\begin{pmatrix}5\\29\end{pmatrix}$

$\mathbf{r}=\begin{pmatrix}2\\9\end{pmatrix}+\lambda\begin{pmatrix}5\\29\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}-3\\-20\end{pmatrix}+\lambda\begin{pmatrix}5\\29\end{pmatrix}$

16) f)

$\begin{pmatrix}-4\\-1\end{pmatrix}-\begin{pmatrix}-10\\-7\end{pmatrix}=\begin{pmatrix}6\\6\end{pmatrix}$

$\mathbf{r}=\begin{pmatrix}-4\\-1\end{pmatrix}+\lambda\begin{pmatrix}6\\6\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}-10\\-7\end{pmatrix}+\lambda\begin{pmatrix}6\\6\end{pmatrix}$

or $\mathbf{r}=\begin{pmatrix}-4\\-1\end{pmatrix}+\lambda\begin{pmatrix}1\\1\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}-10\\-7\end{pmatrix}+\lambda\begin{pmatrix}1\\1\end{pmatrix}$

16) g)

$\begin{pmatrix}k-2\\2k\end{pmatrix}-\begin{pmatrix}k\\k+1\end{pmatrix}=\begin{pmatrix}-2\\k-1\end{pmatrix}$

$\mathbf{r}=\begin{pmatrix}k\\k+1\end{pmatrix}+\lambda\begin{pmatrix}-2\\k-1\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}k-2\\2k\end{pmatrix}+\lambda\begin{pmatrix}-2\\k-1\end{pmatrix}$

or $\mathbf{r}=\begin{pmatrix}k\\k+1\end{pmatrix}+\lambda\begin{pmatrix}2\\1-k\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}k-2\\2k\end{pmatrix}+\lambda\begin{pmatrix}2\\1-k\end{pmatrix}$

17)

$3=5-\lambda\Rightarrow\lambda=2$

$2=-2+2\lambda\Rightarrow\lambda=2$

Consistent values of λ

$\therefore (3,2)$ does lie on the line

18)

$3=-4+2\lambda\Rightarrow\lambda=\frac{7}{2}$

$-2=2+\lambda\Rightarrow\lambda=-4$

Inconsistent values of λ

$\therefore (3,-2)$ does not lie on the line

19)

$-15=-7-2\lambda\Rightarrow\lambda=4$

$15=3+3\lambda\Rightarrow\lambda=4$

Consistent values of λ

$\therefore (-15,15)$ lies on the line

20)

$4 = -3 + k\lambda$ (1)

$-2 = 9 - 2\lambda \Rightarrow \lambda = \frac{11}{2}$ (2)

Substitute (2) into equation (1):

$4 = -3 + k \times \frac{11}{2} \Rightarrow k = \frac{14}{11}$

21)

$-3 = 4 + 3\lambda \Rightarrow \lambda = -\frac{7}{3}$ (1)

$-8 = -1 + k\lambda$ (2)

Substitute (1) into equation (2):

$-8 = -1 + k \times \left(-\frac{7}{3}\right) \Rightarrow k = 3$

22) a)

$4 = k(-5) \Rightarrow k = -\frac{4}{5}$

$1 = k(2) \Rightarrow k = \frac{1}{2}$

Inconsistent values of k, so the lines are not parallel (or coincident).

$-2 + 4\lambda = 1 - 5\mu \Rightarrow 4\lambda + 5\mu = 3$ (1)

$-3 + \lambda = 1 + 2\mu \Rightarrow \lambda - 2\mu = 4$ (2)

Solve (1) and (2): $\lambda = 2$ and $\mu = -1$

Substitute $\lambda = 2$ into l_1:

The lines intersect at $(6, -1)$

22) b)

$3 = k(-3) \Rightarrow k = -1$

$-5 = k(5) \Rightarrow k = -1$

Consistent values of k, so the lines are parallel (or coincident).

Check if $(4, 6)$ lies on l_2:

$4 = -5 - 3\mu \Rightarrow \mu = -3$

$6 = 21 + 5\mu \Rightarrow \mu = -3$

Consistent values of μ, so $(4, 6)$ lies on l_2

As l_1 and l_2 have the same direction vector, and contain the same point, they must be coincident.

22) c)

$-2 = k(4) \Rightarrow k = -\frac{1}{2}$

$9 = k(-18) \Rightarrow k = -\frac{1}{2}$

Consistent values of k, so the lines are parallel (or coincident).

Check if $(6, -1)$ lies on l_2:

$6 = 1 + 4\mu \Rightarrow \mu = \frac{5}{4}$

$-1 = 25 - 18\mu \Rightarrow \mu = \frac{13}{9}$

Inconsistent values of μ, so $(6, -1)$ does not lie on l_2

As l_1 and l_2 have the same direction vector, but do not share the same point, they must be parallel and not coincident.

22) d)

$5 = k(5) \Rightarrow k = 1$

$-6 = k(6) \Rightarrow k = -1$

Inconsistent values of k, so the lines are not parallel (or coincident).

$3 + 5\lambda = 4 + 5\mu \Rightarrow 5\lambda - 5\mu = 1$ (1)

$-4 - 6\lambda = -3 + 6\mu \Rightarrow -6\lambda - 6\mu = 1$ (2)

Solve (1) and (2): $\lambda = \frac{1}{60}$ and $\mu = -\frac{11}{60}$

Substitute $\lambda = \frac{1}{60}$ into l_1:

The lines intersect at $\left(\frac{37}{12}, -\frac{41}{10}\right)$

23)

$\begin{pmatrix} -12 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 20 \end{pmatrix} = (-12) \times 5 + 3 \times 20 = -60 + 60 = 0$

$\therefore l_1$ and l_2 are perpendicular

24) a)

$\begin{pmatrix} 5 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ 3 \end{pmatrix} = 5 \times (-5) + 3 \times 3 = -25 + 9 = -16 \neq 0$

$\therefore l_1$ and l_2 are not perpendicular

24) b)

$\begin{pmatrix} 2 \\ -7 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 7 \end{pmatrix} = 2 \times (-2) + (-7) \times 7 = -4 - 49 = -53 \neq 0$

$\therefore l_1$ and l_2 are not perpendicular

24) c)

$\begin{pmatrix} 9 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 9 \end{pmatrix} = 9 \times 2 + 2 \times 9 = 18 + 18 = 36 \neq 0$

$\therefore l_1$ and l_2 are not perpendicular

24) d)

$\begin{pmatrix} 3 \\ -11 \end{pmatrix} \cdot \begin{pmatrix} 11 \\ 3 \end{pmatrix} = 3 \times 11 + (-11) \times 3 = 33 - 33 = 0$

$\therefore l_1$ and l_2 are perpendicular

25) a)

$\begin{pmatrix} k \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 9 \end{pmatrix} = 0 \Rightarrow k \times 4 + (-3) \times 9 = 0$

$\Rightarrow 4k - 27 = 0 \Rightarrow k = \frac{27}{4}$

25) b)

$1 + \frac{27}{4}\lambda = 2 + 4\mu \Rightarrow \frac{27}{4}\lambda - 4\mu = 1$ (1)

$-100 - 3\lambda = 557 + 9\mu \Rightarrow -3\lambda - 9\mu = 657$ (2)

Solve (1) and (2): $\lambda = -36$ and $\mu = -61$

Substitute $\lambda = -36$ into l_1:

The lines intersect at $(-242, 8)$

So the position vector of the point of intersection is $\begin{pmatrix} -242 \\ 8 \end{pmatrix}$

26) a)

$\begin{pmatrix} -5 \\ k \end{pmatrix} \cdot \begin{pmatrix} 20 \\ 20 \end{pmatrix} = 0 \Rightarrow (-5) \times 20 + k \times 20 = 0$

$\Rightarrow -100 + 20k = 0 \Rightarrow k = 5$

26) b)

$4 - 5\lambda = -2 - \mu \Rightarrow -5\lambda + \mu = -6$ (1)

$8 + 4\lambda = 14 + 20\mu \Rightarrow 5\lambda - 20\mu = 6$ (2)

Solve (1) and (2): $\lambda = \frac{6}{5}$ and $\mu = 0$

Substitute $\lambda = \frac{6}{5}$ into l_1:

The lines intersect at $P(-2, 14)$

26) c)

$\overrightarrow{OP} = \begin{pmatrix} -2 \\ 14 \end{pmatrix}$

$\left|\overrightarrow{OP}\right| = \sqrt{(-2)^2 + 14^2} = \sqrt{200} = 10\sqrt{2}$

Let $\hat{\mathbf{p}}$ be a unit vector in the direction of $\overrightarrow{OP}$

$$\hat{\mathbf{p}} = \frac{\overrightarrow{OP}}{\left|\overrightarrow{OP}\right|} = \frac{1}{10\sqrt{2}}\begin{pmatrix} -2 \\ 14 \end{pmatrix} = \frac{\sqrt{2}}{20}\begin{pmatrix} -2 \\ 14 \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{10} \\ \frac{7\sqrt{2}}{10} \end{pmatrix}$$

27)

If l_1 and l_2 intersect and are not coincident, then they cannot be parallel.

If they were parallel then $\begin{pmatrix} 15 \\ k \end{pmatrix} = p\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ for some constant p

$15 = 3p \Rightarrow p = 5$

$k = 5 \times 2 = 10$

So k can be any real value except 10, i.e. $k \neq 10$

28) a)

$5 + 3\lambda = 4 + 10\mu \Rightarrow 3\lambda - 10\mu = -1$ (1)

$420 - 10\lambda = 120 - 3\mu \Rightarrow -10\lambda + 3\mu = -300$ (2)

Solve (1) and (2): $\lambda = 33$ and $\mu = 10$

Substitute $\lambda = 33$ into l_1:

The lines intersect at $(104, 90)$

28) b)

$\begin{pmatrix}104\\90\end{pmatrix}-\begin{pmatrix}90\\100\end{pmatrix}=\begin{pmatrix}14\\-10\end{pmatrix}$

$\therefore l_3: \mathbf{r}=\begin{pmatrix}90\\100\end{pmatrix}+\gamma\begin{pmatrix}14\\-10\end{pmatrix}$ or $l_3: \mathbf{r}=\begin{pmatrix}90\\100\end{pmatrix}+\gamma\begin{pmatrix}7\\-5\end{pmatrix}$

29) a)

$\cos\theta=\frac{\mathbf{d}_1\cdot\mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|}=\frac{\begin{pmatrix}2\\5\end{pmatrix}\cdot\begin{pmatrix}3\\9\end{pmatrix}}{\left|\begin{pmatrix}2\\5\end{pmatrix}\right|\left|\begin{pmatrix}3\\9\end{pmatrix}\right|}=\frac{2\times3+5\times9}{\sqrt{2^2+5^2}\times\sqrt{3^2+9^2}}=\frac{51}{\sqrt{29}\sqrt{90}}=\frac{17\sqrt{290}}{290}$

$\theta=\arccos\left(\frac{17\sqrt{290}}{290}\right)=3.37°$ (1 DP)

29) b)

$\cos\theta=\frac{\mathbf{d}_1\cdot\mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|}=\frac{\begin{pmatrix}3\\4\end{pmatrix}\cdot\begin{pmatrix}-5\\11\end{pmatrix}}{\left|\begin{pmatrix}3\\4\end{pmatrix}\right|\left|\begin{pmatrix}-5\\11\end{pmatrix}\right|}=\frac{3\times(-5)+4\times11}{\sqrt{3^2+4^2}\times\sqrt{(-5)^2+11^2}}=\frac{29}{5\sqrt{146}}=\frac{29\sqrt{146}}{730}$

$\theta=\arccos\left(\frac{29\sqrt{146}}{730}\right)=61.3°$ (1 DP)

29) c)

$\cos\theta=\frac{\mathbf{d}_1\cdot\mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|}=\frac{\begin{pmatrix}-2\\3\end{pmatrix}\cdot\begin{pmatrix}5\\1\end{pmatrix}}{\left|\begin{pmatrix}-2\\3\end{pmatrix}\right|\left|\begin{pmatrix}5\\1\end{pmatrix}\right|}=\frac{(-2)\times5+3\times1}{\sqrt{(-2)^2+3^2}\times\sqrt{5^2+1^2}}=-\frac{7}{\sqrt{13}\sqrt{26}}=-\frac{7\sqrt{2}}{26}$

$\theta=\arccos\left(-\frac{7\sqrt{2}}{26}\right)=112.380\ldots°$

Acute angle $=180°-112.380\ldots°=67.6°$ (1 DP)

29) d)

$\cos\theta=\frac{\mathbf{d}_1\cdot\mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|}=\frac{\begin{pmatrix}-6\\13\end{pmatrix}\cdot\begin{pmatrix}2\\-3\end{pmatrix}}{\left|\begin{pmatrix}-6\\13\end{pmatrix}\right|\left|\begin{pmatrix}2\\-3\end{pmatrix}\right|}=\frac{(-6)\times2+13\times(-3)}{\sqrt{(-6)^2+13^2}\times\sqrt{2^2+(-3)^2}}=-\frac{51}{\sqrt{205}\sqrt{13}}=-\frac{51\sqrt{2665}}{2665}$

$\theta=\arccos\left(-\frac{51\sqrt{2665}}{2665}\right)=171.085\ldots°$

Acute angle $=180°-171.085\ldots°=8.9°$ (1 DP)

30) a)

$\cos\theta=\frac{\mathbf{d}_1\cdot\mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|}=\frac{\begin{pmatrix}3\\-5\end{pmatrix}\cdot\begin{pmatrix}1\\0\end{pmatrix}}{\left|\begin{pmatrix}3\\-5\end{pmatrix}\right|\left|\begin{pmatrix}1\\0\end{pmatrix}\right|}=\frac{3\times1+(-5)\times0}{\sqrt{3^2+(-5)^2}\times\sqrt{1^2+0^2}}=\frac{3}{\sqrt{34}\sqrt{1}}=\frac{3\sqrt{34}}{34}$

$\theta=\arccos\left(\frac{3\sqrt{34}}{34}\right)=59.0°$ (1 DP)

30) b)

$\cos\theta = \frac{\mathbf{d_1}\cdot\mathbf{d_2}}{|\mathbf{d_1}||\mathbf{d_2}|} = \frac{\binom{3}{-5}\cdot\binom{0}{1}}{\left|\binom{3}{-5}\right|\left|\binom{0}{1}\right|} = \frac{3\times0+(-5)\times1}{\sqrt{3^2+(-5)^2}\times\sqrt{0^2+1^2}} = -\frac{5}{\sqrt{34}\sqrt{1}} = -\frac{5\sqrt{34}}{34}$

$\theta = \arccos\left(-\frac{5\sqrt{34}}{34}\right) = 149.036\ldots°$

Acute angle $= 180° - 149.036\ldots° = 31.0°$ (1 DP)

31)

$\cos\theta = \frac{\mathbf{d_1}\cdot\mathbf{d_2}}{|\mathbf{d_1}||\mathbf{d_2}|} = \frac{\binom{4}{-1}\cdot\binom{3}{4}}{\left|\binom{4}{-1}\right|\left|\binom{3}{4}\right|} = \frac{4\times3+(-1)\times4}{\sqrt{4^2+(-1)^2}\times\sqrt{3^2+4^2}} = \frac{8}{\sqrt{17}\times5} = \frac{8\sqrt{17}}{85}$

$\theta = \arccos\left(\frac{8\sqrt{17}}{85}\right) = 67.166\ldots°$

Obtuse angle $= 180° - 67.166\ldots° = 112.8°$ (1 DP)

32)

$\begin{pmatrix}3 & 2\\ -1 & 5\end{pmatrix}\begin{pmatrix}3+\lambda\\ -2+4\lambda\end{pmatrix} = \begin{pmatrix}3(3+\lambda)+2(-2+4\lambda)\\ -1(3+\lambda)+5(-2+4\lambda)\end{pmatrix}$

$= \begin{pmatrix}5+11\lambda\\ -13+19\lambda\end{pmatrix} = \begin{pmatrix}5\\ -13\end{pmatrix} + \lambda\begin{pmatrix}11\\ 19\end{pmatrix}$

$\therefore l_2: \mathbf{r} = \begin{pmatrix}5\\ -13\end{pmatrix} + \lambda\begin{pmatrix}11\\ 19\end{pmatrix}$

33)

$\begin{pmatrix}4 & 1\\ 2 & -3\end{pmatrix}\begin{pmatrix}6+2\lambda\\ -3-7\lambda\end{pmatrix} = \begin{pmatrix}4(6+2\lambda)+1(-3-7\lambda)\\ 2(6+2\lambda)-3(-3-7\lambda)\end{pmatrix}$

$= \begin{pmatrix}21+\lambda\\ 21+25\lambda\end{pmatrix} = \begin{pmatrix}21\\ 21\end{pmatrix} + \lambda\begin{pmatrix}1\\ 25\end{pmatrix}$

$\therefore l_2: \mathbf{r} = \begin{pmatrix}21\\ 21\end{pmatrix} + \lambda\begin{pmatrix}1\\ 25\end{pmatrix}$

34) a)

$\begin{pmatrix}5 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}-9+5\lambda\\ 4-8\lambda\end{pmatrix} = \begin{pmatrix}5(-9+5\lambda)+0(4-8\lambda)\\ 0(-9+5\lambda)+1(4-8\lambda)\end{pmatrix}$

$= \begin{pmatrix}-45+25\lambda\\ 4-8\lambda\end{pmatrix} = \begin{pmatrix}-45\\ 4\end{pmatrix} + \lambda\begin{pmatrix}25\\ -8\end{pmatrix}$

$\therefore l_2: \mathbf{r} = \begin{pmatrix}-45\\ 4\end{pmatrix} + \lambda\begin{pmatrix}25\\ -8\end{pmatrix}$

34) b)

$$\begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}\begin{pmatrix}-9+5\lambda\\ 4-8\lambda\end{pmatrix}=\begin{pmatrix}2(-9+5\lambda)+0(4-8\lambda)\\ 0(-9+5\lambda)+2(4-8\lambda)\end{pmatrix}$$

$$=\begin{pmatrix}-18+10\lambda\\ 8-16\lambda\end{pmatrix}=\begin{pmatrix}-18\\ 8\end{pmatrix}+\lambda\begin{pmatrix}10\\ -16\end{pmatrix}$$

$$\therefore l_2:\mathbf{r}=\begin{pmatrix}-18\\ 8\end{pmatrix}+\lambda\begin{pmatrix}10\\ -16\end{pmatrix}\text{ or }l_2:\mathbf{r}=\begin{pmatrix}-18\\ 8\end{pmatrix}+\lambda\begin{pmatrix}5\\ -8\end{pmatrix}$$

34) c)

$$\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}-9+5\lambda\\ 4-8\lambda\end{pmatrix}=\begin{pmatrix}1(-9+5\lambda)+0(4-8\lambda)\\ 0(-9+5\lambda)-1(4-8\lambda)\end{pmatrix}$$

$$=\begin{pmatrix}-9+5\lambda\\ -4+8\lambda\end{pmatrix}=\begin{pmatrix}-9\\ -4\end{pmatrix}+\lambda\begin{pmatrix}5\\ 8\end{pmatrix}$$

$$\therefore l_2:\mathbf{r}=\begin{pmatrix}-9\\ -4\end{pmatrix}+\lambda\begin{pmatrix}5\\ 8\end{pmatrix}$$

34) d)

$$\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}-9+5\lambda\\ 4-8\lambda\end{pmatrix}=\begin{pmatrix}-1(-9+5\lambda)+0(4-8\lambda)\\ 0(-9+5\lambda)+1(4-8\lambda)\end{pmatrix}$$

$$=\begin{pmatrix}9-5\lambda\\ 4-8\lambda\end{pmatrix}=\begin{pmatrix}9\\ 4\end{pmatrix}+\lambda\begin{pmatrix}-5\\ -8\end{pmatrix}$$

$$\therefore l_2:\mathbf{r}=\begin{pmatrix}9\\ 4\end{pmatrix}+\lambda\begin{pmatrix}-5\\ -8\end{pmatrix}\text{ or }l_2:\mathbf{r}=\begin{pmatrix}9\\ 4\end{pmatrix}+\lambda\begin{pmatrix}5\\ 8\end{pmatrix}$$

34) e)

$$\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}-9+5\lambda\\ 4-8\lambda\end{pmatrix}=\begin{pmatrix}0(-9+5\lambda)+1(4-8\lambda)\\ 1(-9+5\lambda)+0(4-8\lambda)\end{pmatrix}$$

$$=\begin{pmatrix}4-8\lambda\\ -9+5\lambda\end{pmatrix}=\begin{pmatrix}4\\ -9\end{pmatrix}+\lambda\begin{pmatrix}-8\\ 5\end{pmatrix}$$

$$\therefore l_2:\mathbf{r}=\begin{pmatrix}4\\ -9\end{pmatrix}+\lambda\begin{pmatrix}-8\\ 5\end{pmatrix}$$

34) f)

$$\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}-9+5\lambda\\ 4-8\lambda\end{pmatrix}=\begin{pmatrix}-1(-9+5\lambda)+0(4-8\lambda)\\ 0(-9+5\lambda)-1(4-8\lambda)\end{pmatrix}$$

$$=\begin{pmatrix}9-5\lambda\\ -4+8\lambda\end{pmatrix}=\begin{pmatrix}9\\ -4\end{pmatrix}+\lambda\begin{pmatrix}-5\\ 8\end{pmatrix}$$

$$\therefore l_2:\mathbf{r}=\begin{pmatrix}9\\ -4\end{pmatrix}+\lambda\begin{pmatrix}-5\\ 8\end{pmatrix}$$

35)

Using a non-Vector method:

The line passing through $(0,0)$ which is perpendicular to $y = 2x + 5$ is $y = -\frac{1}{2}x$

$y = 2x + 5$ and $y = -\frac{1}{2}x$ intersect at $(-2, 1)$

Distance between $(0,0)$ and $(-2,1)$ is $\sqrt{(-2-0)^2 + (1-0)^2} = \sqrt{5}$ units

Alternative Method (using Vectors):

Let l be the line $\mathbf{r} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} \lambda \\ 5 + 2\lambda \end{pmatrix}$

$\overrightarrow{OP} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} \lambda \\ 5 + 2\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 0$

$\Rightarrow \lambda \times 1 + (5 + 2\lambda) \times 2 = 0$

$\Rightarrow \lambda + 10 + 4\lambda = 0$

$\Rightarrow 5\lambda + 10 = 0$

$\lambda = -2$

So $\overrightarrow{OP} = \begin{pmatrix} -2 \\ 5 + 2(-2) \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

So $\left|\overrightarrow{OP}\right| = \sqrt{(-2)^2 + 1^2} = \sqrt{5}$ units

Alternative Method (using the Vector Product):

Convert to a 3D problem:

Let $\mathbf{p} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \times \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix}\right|}{\left|\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}\right|}$

$\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 0 \\ 0 & -5 & 0 \end{vmatrix}$

$= (2 \times 0 - 0 \times (-5))\mathbf{i} - (1 \times 0 - 0 \times 0)\mathbf{j} + (1 \times (-5) - 2 \times 0)\mathbf{k}$

$= 0\mathbf{i} + 0\mathbf{j} - 5\mathbf{k}$

$$\therefore h = \frac{\left|\begin{pmatrix} 0 \\ 0 \\ -5 \end{pmatrix}\right|}{\left|\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}\right|} = \frac{\sqrt{0^2+0^2+(-5)^2}}{\sqrt{1^2+2^2+0^2}} = \frac{5}{\sqrt{5}} = \sqrt{5} \text{ units}$$

36)

Using a non-Vector method:

The line passing through $(3,-5)$ which is perpendicular to $y=8x+1$ is

$y+5=-\frac{1}{8}(x-3) \Rightarrow y=-\frac{1}{8}x-\frac{37}{8}$

$y=8x+1$ and $y=-\frac{1}{8}x-\frac{37}{8}$ intersect at $\left(-\frac{9}{13},-\frac{59}{13}\right)$

Distance between $(3,-5)$ and $\left(-\frac{9}{13},-\frac{59}{13}\right)$ is $\sqrt{\left(-\frac{9}{13}-3\right)^2+\left(-\frac{59}{13}+5\right)^2}$

$=\sqrt{\frac{180}{13}}=\frac{6\sqrt{65}}{13}$ units

Alternative Method (using Vectors):

Let A be the point with coordinates $(3,-5)$

Let l be the line $\mathbf{r}=\begin{pmatrix}0\\1\end{pmatrix}+\lambda\begin{pmatrix}1\\8\end{pmatrix}$

Let P be a point on the line, so $\overrightarrow{OP}=\begin{pmatrix}\lambda\\1+8\lambda\end{pmatrix}$

$\overrightarrow{AP}=\overrightarrow{OP}-\overrightarrow{OA}=\begin{pmatrix}\lambda\\1+8\lambda\end{pmatrix}-\begin{pmatrix}3\\-5\end{pmatrix}=\begin{pmatrix}\lambda-3\\6+8\lambda\end{pmatrix}$

$\overrightarrow{AP}\cdot\begin{pmatrix}1\\8\end{pmatrix}=0 \Rightarrow \begin{pmatrix}\lambda-3\\6+8\lambda\end{pmatrix}\cdot\begin{pmatrix}1\\8\end{pmatrix}=0$

$\Rightarrow(\lambda-3)\times 1+(6+8\lambda)\times 8=0$

$\Rightarrow \lambda-3+48+64\lambda=0$

$\Rightarrow 65\lambda+45=0$

$\lambda=-\frac{9}{13}$

So $\overrightarrow{AP}=\begin{pmatrix}\left(-\frac{9}{13}\right)-3\\6+8\left(-\frac{9}{13}\right)\end{pmatrix}=\begin{pmatrix}-\frac{48}{13}\\\frac{6}{13}\end{pmatrix}$

So $\left|\overrightarrow{AP}\right|=\sqrt{\left(-\frac{48}{13}\right)^2+\left(\frac{6}{13}\right)^2}=\frac{6\sqrt{65}}{13}$ units

Alternative Method (using the Vector Product):

Convert to a 3D problem:

Let $\mathbf{p}=\begin{pmatrix}3\\-5\\0\end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a}=\begin{pmatrix}0\\1\\0\end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 1 \\ 8 \\ 0 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left| \begin{pmatrix} 1 \\ 8 \\ 0 \end{pmatrix} \times \left(\begin{pmatrix} 3 \\ -5 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right) \right|}{\left| \begin{pmatrix} 1 \\ 8 \\ 0 \end{pmatrix} \right|} = \frac{\left| \begin{pmatrix} 1 \\ 8 \\ 0 \end{pmatrix} \times \begin{pmatrix} 3 \\ -6 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} 1 \\ 8 \\ 0 \end{pmatrix} \right|}$

$$\begin{pmatrix} 1 \\ 8 \\ 0 \end{pmatrix} \times \begin{pmatrix} 3 \\ -6 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 8 & 0 \\ 3 & -6 & 0 \end{vmatrix}$$

$$= \big(8 \times 0 - 0 \times (-6)\big)\mathbf{i} - (1 \times 0 - 0 \times 3)\mathbf{j} + (1 \times (-6) - 8 \times 3)\mathbf{k}$$

$$= 0\mathbf{i} + 0\mathbf{j} - 30\mathbf{k}$$

$\therefore h = \frac{\left| \begin{pmatrix} 0 \\ 0 \\ -30 \end{pmatrix} \right|}{\left| \begin{pmatrix} 1 \\ 8 \\ 0 \end{pmatrix} \right|} = \frac{\sqrt{0^2 + 0^2 + (-30)^2}}{\sqrt{1^2 + 8^2 + 0^2}} = \frac{30}{\sqrt{65}} = \frac{6\sqrt{65}}{13}$ units

37)

Using a non-Vector method:

The line has cartesian equation $y - 0 = \frac{1}{2}(x + 1) = y = \frac{1}{2}x + \frac{1}{2}$

The line passing through $(4, -2)$ which is perpendicular to $y = \frac{1}{2}x + \frac{1}{2}$ is

$y + 2 = -2(x - 4) \Rightarrow y = -2x + 6$

$y = \frac{1}{2}x + \frac{1}{2}$ and $y = -2x + 6$ intersect at $\left(\frac{11}{5}, \frac{8}{5}\right)$

Distance between $\left(\frac{11}{5}, \frac{8}{5}\right)$ and $(4, -2)$ is $\sqrt{\left(4 - \frac{11}{5}\right)^2 + \left(-2 - \frac{8}{5}\right)^2}$

$= \sqrt{\frac{81}{5}} = \frac{9\sqrt{5}}{5}$ units

Alternative Method (using Vectors):

Let A be the point with coordinates $(4, -2)$

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} -1 + 2\lambda \\ \lambda \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} -1 + 2\lambda \\ \lambda \end{pmatrix} - \begin{pmatrix} 4 \\ -2 \end{pmatrix} = \begin{pmatrix} -5 + 2\lambda \\ \lambda + 2 \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} -5 + 2\lambda \\ \lambda + 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 0$

$\Rightarrow (-5 + 2\lambda) \times 2 + (\lambda + 2) \times 1 = 0$

$\Rightarrow -10 + 4\lambda + \lambda + 2 = 0$

$\Rightarrow 5\lambda - 8 = 0$

$\lambda = \frac{8}{5}$

So $\overrightarrow{AP} = \begin{pmatrix} -5 + 2\left(\frac{8}{5}\right) \\ \left(\frac{8}{5}\right) + 2 \end{pmatrix} = \begin{pmatrix} -\frac{9}{5} \\ \frac{18}{5} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{\left(-\frac{9}{5}\right)^2 + \left(\frac{18}{5}\right)^2} = \frac{9\sqrt{5}}{5}$ units

Alternative Method (using the Vector Product):

Convert to a 3D problem:

Let $\mathbf{p} = \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \left(\begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} \right) \right|}{\left| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right|} = \frac{\left| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 5 \\ -2 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right|}$

$$\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 5 \\ -2 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 0 \\ 5 & -2 & 0 \end{vmatrix}$$

$$= \big(1 \times 0 - 0 \times (-2)\big)\mathbf{i} - (2 \times 0 - 0 \times 5)\mathbf{j} + (2 \times (-2) - 1 \times 5)\mathbf{k}$$

$$= 0\mathbf{i} + 0\mathbf{j} - 9\mathbf{k}$$

$\therefore h = \frac{\left| \begin{pmatrix} 0 \\ 0 \\ -9 \end{pmatrix} \right|}{\left| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right|} = \frac{\sqrt{0^2 + 0^2 + (-9)^2}}{\sqrt{2^2 + 1^2 + 0^2}} = \frac{9}{\sqrt{5}} = \frac{9\sqrt{5}}{5}$ units

38)

Using a non-Vector method:

The line has cartesian equation $y+2=-\frac{3}{5}(x-4)=y=-\frac{3}{5}x+\frac{2}{5}$

The line passing through $(1,k)$ which is perpendicular to $y=-\frac{3}{5}x+\frac{2}{5}$ is

$y-k=\frac{5}{3}(x-1)\Rightarrow y=\frac{5}{3}x+k-\frac{5}{3}$

$y=-\frac{3}{5}x+\frac{2}{5}$ and $y=\frac{5}{3}x+k-\frac{5}{3}$ intersect when:

$-\frac{3}{5}x+\frac{2}{5}=\frac{5}{3}x+k-\frac{5}{3}\Rightarrow\frac{34}{15}x=\frac{31}{15}-k\Rightarrow x=\frac{31}{34}-\frac{15}{34}k$

$\therefore y=-\frac{3}{5}\left(\frac{31}{34}-\frac{15}{34}k\right)+\frac{2}{5}=\frac{9}{34}k-\frac{5}{34}$

The lines intersect at $\left(\frac{31}{34}-\frac{15}{34}k,\frac{9}{34}k-\frac{5}{34}\right)$

Distance between $\left(\frac{31}{34}-\frac{15}{34}k,\frac{9}{34}k-\frac{5}{34}\right)$ and $(1,k)$ is 6664 unit:

$$\sqrt{\left(1-\left(\frac{31}{34}-\frac{15}{34}k\right)\right)^2+\left(k-\left(\frac{9}{34}k-\frac{5}{34}\right)\right)^2}=6664$$

$\Rightarrow\left(\frac{15}{34}k+\frac{3}{34}\right)^2+\left(\frac{25k}{34}+\frac{5}{34}\right)^2=6664$

$\Rightarrow\left(\frac{1}{34}\right)^2(15k+3)^2+\left(\frac{1}{34}\right)^2(25k+5)^2=6664$

$\Rightarrow(15k+3)^2+(25k+5)^2=7703584$

$\Rightarrow 225k^2+90k+9+625k^2+250k+25=7703584$

$\Rightarrow 850k^2+340k-7703550=0$

$\Rightarrow 5k^2+2k-45315=0$

$\Rightarrow(5k+477)(k-95)=0$

$\Rightarrow k=-\frac{477}{5}$ or $k=95$

As k is an integer, $k=95$

Alternative Method, (using Vectors):

Let A be the point with coordinates $(1,k)$

Let P be a point on the line, so $\overrightarrow{OP}=\begin{pmatrix}4-5\lambda\\-2+3\lambda\end{pmatrix}$

$\overrightarrow{AP}=\overrightarrow{OP}-\overrightarrow{OA}=\begin{pmatrix}4-5\lambda\\-2+3\lambda\end{pmatrix}-\begin{pmatrix}1\\k\end{pmatrix}=\begin{pmatrix}3-5\lambda\\-2-k+3\lambda\end{pmatrix}$

$\overrightarrow{AP}\cdot\begin{pmatrix}-5\\3\end{pmatrix}=0\Rightarrow\begin{pmatrix}3-5\lambda\\-2-k+3\lambda\end{pmatrix}\cdot\begin{pmatrix}-5\\3\end{pmatrix}=0$

$\Rightarrow(3-5\lambda)\times(-5)+(-2-k+3\lambda)\times 3=0$

$\Rightarrow -15+25\lambda-6-3k+9\lambda=0$

$\Rightarrow 34\lambda - 21 - 3k = 0$

$\lambda = \frac{21+3k}{34}$

So $\overrightarrow{AP} = \begin{pmatrix} 3 - 5\left(\frac{21+3k}{34}\right) \\ -2 - k + 3\left(\frac{21+3k}{34}\right) \end{pmatrix} = \begin{pmatrix} -\frac{15}{34}k - \frac{3}{34} \\ -\frac{25}{34}k - \frac{5}{34} \end{pmatrix}$

$\left|\overrightarrow{AP}\right| = 14\sqrt{34} \Rightarrow \sqrt{\left(-\frac{15}{34}k - \frac{3}{34}\right)^2 + \left(-\frac{25}{34}k - \frac{5}{34}\right)^2} = 14\sqrt{34}$

$\Rightarrow \left(\frac{1}{34}\right)^2 (15k+3)^2 + \left(\frac{1}{34}\right)^2 (25k+5)^2 = 6664$

$\Rightarrow (15k+3)^2 + (25k+5)^2 = 7703584$

$\Rightarrow 225k^2 + 90k + 9 + 625k^2 + 250k + 25 = 7703584$

$\Rightarrow 850k^2 + 340k - 7703550 = 0$

$\Rightarrow 5k^2 + 2k - 45315 = 0$

$\Rightarrow (5k + 477)(k - 95) = 0$

$\Rightarrow k = -\frac{477}{5}$ or $k = 95$

As k is an integer, $k = 95$

Alternative Method (using the Vector Product):

Convert to a 3D problem:

Let $\mathbf{p} = \begin{pmatrix} 1 \\ k \\ 0 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix} \times \left(\begin{pmatrix} 1 \\ k \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix} \times \begin{pmatrix} -3 \\ k+2 \\ 0 \end{pmatrix}\right|}{\left|\begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix}\right|}$

$\begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix} \times \begin{pmatrix} -3 \\ k+2 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -5 & 3 & 0 \\ -3 & k+2 & 0 \end{vmatrix}$

$= \big(3 \times 0 - 0 \times (k+2)\big)\mathbf{i} - \big((-5) \times 0 - 0 \times (-3)\big)\mathbf{j} + \big((-5) \times (k+2) - 3 \times (-3)\big)\mathbf{k}$

$= 0\mathbf{i} + 0\mathbf{j} + (-5k - 1)\mathbf{k}$

$\therefore h = 14\sqrt{34} \Rightarrow \frac{\left|\begin{pmatrix} 0 \\ 0 \\ -5k-1 \end{pmatrix}\right|}{\left|\begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix}\right|} = 14\sqrt{34} \Rightarrow \frac{\sqrt{0^2+0^2+(-5k-1)^2}}{\sqrt{(-5)^2+3^2+0^2}} = 14\sqrt{34}$

$\Rightarrow \frac{\sqrt{25k^2+10k+1}}{\sqrt{34}} = 14\sqrt{34} \Rightarrow \sqrt{25k^2+10k+1} = 476$

$\Rightarrow 25k^2 + 10k + 1 = 226576$

$\Rightarrow 25k^2 + 10k - 226575 = 0$

$\Rightarrow 5k^2 + 2k - 45315 = 0$

$\Rightarrow (5k + 477)(k - 95) = 0$

$\Rightarrow k = -\frac{477}{5}$ or $k = 95$

As k is an integer, $k = 95$

39)

Using a non-Vector method:

Choose a point on l_1 : $(0, 2)$

The line passing through $(0, 2)$ which is perpendicular to $y = x + 2$ is

$y - 2 = -1(x - 0) \Rightarrow y = -x + 2$

$y = -x + 2$ and $y = x + 8$ intersect at $(-3, 5)$

Distance between $(-3, 5)$ and $(0, 2)$ is $\sqrt{(0 + 3)^2 + (2 - 5)^2}$

$= \sqrt{18} = 3\sqrt{2}$ units

Alternative Method (using Vectors):

Choose a point on l_1 so let A be the point with coordinates $(0, 2)$

Let l_2 be the line $\mathbf{r} = \begin{pmatrix} 0 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} \lambda \\ 8 + \lambda \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} \lambda \\ 8 + \lambda \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} \lambda \\ 6 + \lambda \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} \lambda \\ 6 + \lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0$

$\Rightarrow \lambda \times 1 + (6 + \lambda) \times 1 = 0$

$\Rightarrow \lambda + 6 + \lambda = 0$

$\Rightarrow 2\lambda + 6 = 0$

$\lambda = -3$

So $\overrightarrow{AP} = \begin{pmatrix} -3 \\ 6 + (-3) \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{(-3)^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$ units

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Convert to a 3D problem:

Let $\mathbf{p} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$

Let A be a known point on the line l_2 so that its position vector is $\mathbf{a} = \begin{pmatrix} 0 \\ 8 \\ 0 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

Then $h = \frac{|\mathbf{d}\times(\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix}1\\1\\0\end{pmatrix}\times\left(\begin{pmatrix}0\\2\\0\end{pmatrix}-\begin{pmatrix}0\\8\\0\end{pmatrix}\right)\right|}{\left|\begin{pmatrix}1\\1\\0\end{pmatrix}\right|} = \frac{\left|\begin{pmatrix}1\\1\\0\end{pmatrix}\times\begin{pmatrix}0\\-6\\0\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\1\\0\end{pmatrix}\right|}$

$\begin{pmatrix}1\\1\\0\end{pmatrix}\times\begin{pmatrix}0\\-6\\0\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\1 & 1 & 0\\0 & -6 & 0\end{vmatrix}$

$= \big(1\times 0 - 0\times(-6)\big)\mathbf{i} - (1\times 0 - 0\times 0)\mathbf{j} + (1\times(-6) - 1\times 0)\mathbf{k}$

$= 0\mathbf{i} + 0\mathbf{j} - 6\mathbf{k}$

$\therefore h = \frac{\left|\begin{pmatrix}0\\0\\-6\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\1\\0\end{pmatrix}\right|} = \frac{\sqrt{0^2+0^2+(-6)^2}}{\sqrt{1^2+1^2+0^2}} = \frac{6}{\sqrt{2}} = 3\sqrt{2}$ units

40)

Using a non-Vector method:

Choose a point on l_1 : $(0, -20)$

The line passing through $(0, -20)$ which is perpendicular to $y = 5x + 30$ is

$y + 20 = -\frac{1}{5}(x - 0) \Rightarrow y = -\frac{1}{5}x - 20$

$y = -\frac{1}{5}x - 20$ and $y = 5x + 30$ intersect at $\left(-\frac{125}{13}, -\frac{235}{13}\right)$

Distance between $\left(-\frac{125}{13}, -\frac{235}{13}\right)$ and $(0, -20)$ is $\sqrt{\left(0 + \frac{125}{13}\right)^2 + \left(-20 + \frac{235}{13}\right)^2}$

$= \sqrt{\frac{1250}{13}} = \frac{25\sqrt{26}}{13}$ units

Alternative Method (using Vectors):

Choose a point on l_1 so let A be the point with coordinates $(0, -20)$

Let l_2 be the line $\mathbf{r} = \begin{pmatrix} 0 \\ 30 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \end{pmatrix}$

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} \lambda \\ 30 + 5\lambda \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} \lambda \\ 30 + 5\lambda \end{pmatrix} - \begin{pmatrix} 0 \\ -20 \end{pmatrix} = \begin{pmatrix} \lambda \\ 50 + 5\lambda \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} 1 \\ 5 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} \lambda \\ 50 + 5\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 5 \end{pmatrix} = 0$

$\Rightarrow \lambda \times 1 + (50 + 5\lambda) \times 5 = 0$

$\Rightarrow \lambda + 250 + 25\lambda = 0$

$\Rightarrow 26\lambda + 250 = 0$

$\lambda = -\frac{125}{13}$

So $\overrightarrow{AP} = \begin{pmatrix} \left(-\frac{125}{13}\right) \\ 50 + 5\left(-\frac{125}{13}\right) \end{pmatrix} = \begin{pmatrix} -\frac{125}{13} \\ \frac{25}{13} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{\left(-\frac{125}{13}\right)^2 + \left(\frac{25}{13}\right)^2} = \frac{25\sqrt{26}}{13}$ units

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Convert to a 3D problem:

Let $\mathbf{p} = \begin{pmatrix} 0 \\ -20 \\ 0 \end{pmatrix}$

Let A be a known point on the line l_2 so that its position vector is $\mathbf{a} = \begin{pmatrix} 0 \\ 30 \\ 0 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \left(\begin{pmatrix} 0 \\ -20 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 30 \\ 0 \end{pmatrix} \right) \right|}{\left| \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix} \right|} = \frac{\left| \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -50 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix} \right|}$

$$\begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -50 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 5 & 0 \\ 0 & -50 & 0 \end{vmatrix}$$

$$= \big(5 \times 0 - 0 \times (-50)\big)\mathbf{i} - (1 \times 0 - 0 \times 0)\mathbf{j} + (1 \times (-50) - 5 \times 0)\mathbf{k}$$

$$= 0\mathbf{i} + 0\mathbf{j} - 50\mathbf{k}$$

$\therefore h = \frac{\left| \begin{pmatrix} 0 \\ 0 \\ -50 \end{pmatrix} \right|}{\left| \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix} \right|} = \frac{\sqrt{0^2 + 0^2 + (-50)^2}}{\sqrt{1^2 + 5^2 + 0^2}} = \frac{50}{\sqrt{26}} = \frac{25\sqrt{26}}{13}$ units

41)

Using a non-Vector method:

Choose a point on l_1 : $(0, 2)$

The line passing through $(0, 2)$ which is perpendicular to $y = -1 - 10x$ is

$y - 2 = \frac{1}{10}(x - 0) \Rightarrow y = \frac{1}{10}x + 2$

$y = \frac{1}{10}x + 2$ and $y = -1 - 10x$ intersect at $\left(-\frac{30}{101}, \frac{199}{101}\right)$

Distance between $\left(-\frac{30}{101}, \frac{199}{101}\right)$ and $(0, 2)$ is $\sqrt{\left(0 + \frac{30}{101}\right)^2 + \left(2 - \frac{199}{101}\right)^2}$

$= \sqrt{\frac{9}{101}} = \frac{3\sqrt{101}}{101}$ units

Alternative Method (using Vectors):

Choose a point on l_1 so let A be the point with coordinates $(0, 2)$

Let l_2 be the line $\mathbf{r} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -10 \end{pmatrix}$

Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} \lambda \\ -1 - 10\lambda \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} \lambda \\ -1 - 10\lambda \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} \lambda \\ -3 - 10\lambda \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} 1 \\ -10 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} \lambda \\ -3 - 10\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -10 \end{pmatrix} = 0$

$\Rightarrow \lambda \times 1 + (-3 - 10\lambda) \times (-10) = 0$

$\Rightarrow \lambda + 30 + 100\lambda = 0$

$\Rightarrow 101\lambda + 30 = 0$

$\lambda = -\frac{30}{101}$

So $\overrightarrow{AP} = \begin{pmatrix} \left(-\frac{30}{101}\right) \\ -3 - 10\left(-\frac{30}{101}\right) \end{pmatrix} = \begin{pmatrix} -\frac{30}{101} \\ -\frac{3}{101} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{\left(-\frac{30}{101}\right)^2 + \left(-\frac{3}{101}\right)^2} = \sqrt{\frac{9}{101}} = \frac{3\sqrt{101}}{101}$ units

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Convert to a 3D problem:

Let $\mathbf{p} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$

Let A be a known point on the line l_2 so that its position vector is $\mathbf{a} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 1 \\ -10 \\ 0 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left| \begin{pmatrix} 1 \\ -10 \\ 0 \end{pmatrix} \times \left(\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} \right) \right|}{\left| \begin{pmatrix} 1 \\ -10 \\ 0 \end{pmatrix} \right|} = \frac{\left| \begin{pmatrix} 1 \\ -10 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} 1 \\ -10 \\ 0 \end{pmatrix} \right|}$

$$\begin{pmatrix} 1 \\ -10 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -10 & 0 \\ 0 & 3 & 0 \end{vmatrix}$$

$$= \big((-10) \times 0 - 0 \times 3\big)\mathbf{i} - (1 \times 0 - 0 \times 0)\mathbf{j} + (1 \times 3 - (-10) \times 0)\mathbf{k}$$

$$= 0\mathbf{i} + 0\mathbf{j} + 3\mathbf{k}$$

$\therefore h = \frac{\left| \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} \right|}{\left| \begin{pmatrix} 1 \\ -10 \\ 0 \end{pmatrix} \right|} = \frac{\sqrt{0^2+0^2+3^2}}{\sqrt{1^2+(-10)^2+0^2}} = \frac{3}{\sqrt{101}} = \frac{3\sqrt{101}}{101}$ units

42)

Using a non-Vector method:

l_1 has cartesian equation $y - 1 = \frac{1}{3}(x - 0) \Rightarrow y = \frac{1}{3}x + 1$

l_2 has cartesian equation $y - 0 = \frac{1}{3}(x - 1) \Rightarrow y = \frac{1}{3}x - \frac{1}{3}$

Choose a point on l_1 : $(0, 1)$

The line passing through $(0, 1)$ which is perpendicular to $y = \frac{1}{3}x - \frac{1}{3}$ is

$y - 1 = -3(x - 0) \Rightarrow y = -3x + 1$

$y = -3x + 1$ and $y = \frac{1}{3}x - \frac{1}{3}$ intersect at $\left(\frac{2}{5}, -\frac{1}{5}\right)$

Distance between $\left(\frac{2}{5}, -\frac{1}{5}\right)$ and $(0, 1)$ is $\sqrt{\left(0 - \frac{2}{5}\right)^2 + \left(1 + \frac{1}{5}\right)^2}$

$= \sqrt{\frac{8}{5}} = \frac{2\sqrt{10}}{5}$ units

Alternative Method (using Vectors):

Choose a point on l_1 so let A be the point with coordinates $(0, 1)$

Let P be a point on the line l_2, so $\overrightarrow{OP} = \begin{pmatrix} 1 + 3\mu \\ \mu \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 1 + 3\mu \\ \mu \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 + 3\mu \\ \mu - 1 \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 1 + 3\mu \\ \mu - 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 0$

$\Rightarrow (1 + 3\mu) \times 3 + (\mu - 1) \times 1 = 0$

$\Rightarrow 3 + 9\mu + \mu - 1 = 0$

$\Rightarrow 10\mu + 2 = 0$

$\mu = -\frac{1}{5}$

So $\overrightarrow{AP} = \begin{pmatrix} 1 + 3\left(-\frac{1}{5}\right) \\ \left(-\frac{1}{5}\right) - 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{5} \\ -\frac{6}{5} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{\left(\frac{2}{5}\right)^2 + \left(-\frac{6}{5}\right)^2} = \sqrt{\frac{8}{5}} = \frac{2\sqrt{10}}{5}$ units

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Convert to a 3D problem:

Let $\mathbf{p} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

Let A be a known point on the line l_2 so that its position vector is $\mathbf{a} = \begin{pmatrix}1\\0\\0\end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix}3\\1\\0\end{pmatrix}$

Then $h = \frac{|\mathbf{d}\times(\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix}3\\1\\0\end{pmatrix}\times\left(\begin{pmatrix}0\\1\\0\end{pmatrix}-\begin{pmatrix}1\\0\\0\end{pmatrix}\right)\right|}{\left|\begin{pmatrix}3\\1\\0\end{pmatrix}\right|} = \frac{\left|\begin{pmatrix}3\\1\\0\end{pmatrix}\times\begin{pmatrix}-1\\1\\0\end{pmatrix}\right|}{\left|\begin{pmatrix}3\\1\\0\end{pmatrix}\right|}$

$\begin{pmatrix}3\\1\\0\end{pmatrix}\times\begin{pmatrix}-1\\1\\0\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\3 & 1 & 0\\-1 & 1 & 0\end{vmatrix}$

$= (1\times 0 - 0\times 1)\mathbf{i} - \big(3\times 0 - 0\times(-1)\big)\mathbf{j} + \big(3\times 1 - 1\times(-1)\big)\mathbf{k}$

$= 0\mathbf{i} + 0\mathbf{j} + 4\mathbf{k}$

$\therefore h = \frac{\left|\begin{pmatrix}0\\0\\4\end{pmatrix}\right|}{\left|\begin{pmatrix}3\\1\\0\end{pmatrix}\right|} = \frac{\sqrt{0^2+0^2+4^2}}{\sqrt{3^2+1^2+0^2}} = \frac{4}{\sqrt{10}} = \frac{2\sqrt{10}}{5}$ units

43)

Using a non-Vector method:

l_1 has cartesian equation $y - 8 = -\frac{5}{4}(x - 2) \Rightarrow y = -\frac{5}{4}x + \frac{21}{2}$

l_2 has cartesian equation $y + 4 = -\frac{5}{4}(x + 3) \Rightarrow y = -\frac{5}{4}x - \frac{31}{4}$

Choose a point on l_1 : $\left(0, \frac{21}{2}\right)$

The line passing through $\left(0, \frac{21}{2}\right)$ which is perpendicular to $y = -\frac{5}{4}x + \frac{21}{2}$ is

$y - \frac{21}{2} = \frac{4}{5}(x - 0) \Rightarrow y = \frac{4}{5}x + \frac{21}{2}$

$y = \frac{4}{5}x + \frac{21}{2}$ and $y = -\frac{5}{4}x - \frac{31}{4}$ intersect at $\left(-\frac{365}{41}, \frac{277}{82}\right)$

Distance between $\left(-\frac{365}{41}, \frac{277}{82}\right)$ and $\left(0, \frac{21}{2}\right)$ is $\sqrt{\left(0 + \frac{365}{41}\right)^2 + \left(\frac{21}{2} - \frac{277}{82}\right)^2}$

$= \sqrt{\frac{5329}{41}} = \frac{73\sqrt{41}}{41}$ units

Alternative Method (using Vectors):

Choose a point on l_1 so let A be the point with coordinates $(2, 8)$

Let P be a point on the line l_2, so $\overrightarrow{OP} = \begin{pmatrix} -3 - 4\mu \\ -4 + 5\mu \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} -3 - 4\mu \\ -4 + 5\mu \end{pmatrix} - \begin{pmatrix} 2 \\ 8 \end{pmatrix} = \begin{pmatrix} -5 - 4\mu \\ -12 + 5\mu \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} -4 \\ 5 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} -5 - 4\mu \\ -12 + 5\mu \end{pmatrix} \cdot \begin{pmatrix} -4 \\ 5 \end{pmatrix} = 0$

$\Rightarrow (-5 - 4\mu) \times (-4) + (-12 + 5\mu) \times 5 = 0$

$\Rightarrow 20 + 16\mu - 60 + 25\mu = 0$

$\Rightarrow 41\mu - 40 = 0$

$\Rightarrow \mu = \frac{40}{41}$

So $\overrightarrow{AP} = \begin{pmatrix} -5 - 4\left(\frac{40}{41}\right) \\ -12 + 5\left(\frac{40}{41}\right) \end{pmatrix} = \begin{pmatrix} -\frac{365}{41} \\ -\frac{292}{41} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{\left(-\frac{365}{41}\right)^2 + \left(-\frac{292}{41}\right)^2} = \sqrt{\frac{5329}{41}} = \frac{73\sqrt{41}}{41}$ units

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Convert to a 3D problem:

Let $\mathbf{p} = \begin{pmatrix} 2 \\ 8 \\ 0 \end{pmatrix}$

Let A be a known point on the line l_2 so that its position vector is $\mathbf{a} = \begin{pmatrix} -3 \\ -4 \\ 0 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left| \begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix} \times \left(\begin{pmatrix} 2 \\ 8 \\ 0 \end{pmatrix} - \begin{pmatrix} -3 \\ -4 \\ 0 \end{pmatrix} \right) \right|}{\left| \begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix} \right|} = \frac{\left| \begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix} \times \begin{pmatrix} 5 \\ 12 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix} \right|}$

$$\begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix} \times \begin{pmatrix} 5 \\ 12 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -4 & 5 & 0 \\ 5 & 12 & 0 \end{vmatrix}$$

$$= (5 \times 0 - 0 \times 12)\mathbf{i} - \big((-4) \times 0 - 0 \times 5\big)\mathbf{j} + \big((-4) \times 12 - 5 \times 5\big)\mathbf{k}$$

$$= 0\mathbf{i} + 0\mathbf{j} - 73\mathbf{k}$$

$$\therefore h = \frac{\left| \begin{pmatrix} 0 \\ 0 \\ -73 \end{pmatrix} \right|}{\left| \begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix} \right|} = \frac{\sqrt{0^2 + 0^2 + (-73)^2}}{\sqrt{(-4)^2 + 5^2 + 0^2}} = \frac{73}{\sqrt{41}} = \frac{73\sqrt{41}}{41} \text{ units}$$

44)

Using a non-Vector method:

l_1 has cartesian equation $y - k = \frac{5}{2}(x + 4) \Rightarrow y = \frac{5}{2}x + 10 + k$

l_2 has cartesian equation $y - 9 = \frac{5}{2}(x - 8) \Rightarrow y = \frac{5}{2}x - 11$

Choose a point on l_1 : $(0, 10 + k)$

The line passing through $(0, 10 + k)$ which is perpendicular to $y = \frac{5}{2}x - 11$ is

$y - (10 + k) = -\frac{2}{5}(x - 0) \Rightarrow y = -\frac{2}{5}x + 10 + k$

$y = -\frac{2}{5}x + 10 + k$ and $y = \frac{5}{2}x - 11$ intersect:

$-\frac{2}{5}x + 10 + k = \frac{5}{2}x - 11$

$\Rightarrow \frac{29}{10}x = 21 + k$

$\Rightarrow x = \frac{210}{29} + \frac{10}{29}k$

$\therefore y = \frac{5}{2}\left(\frac{210}{29} + \frac{10}{29}k\right) - 11 = \frac{25}{29}k + \frac{206}{29}$

The lines intersect at $\left(\frac{210}{29} + \frac{10}{29}k, \frac{25}{29}k + \frac{206}{29}\right)$

Distance between $\left(\frac{210}{29} + \frac{10}{29}k, \frac{25}{29}k + \frac{206}{29}\right)$ and $(0, 10 + k)$ is $4\sqrt{29}$:

$$\sqrt{\left(0 - \left(\frac{210}{29} + \frac{10}{29}k\right)\right)^2 + \left((10 + k) - \left(\frac{25}{29}k + \frac{206}{29}\right)\right)^2} = 4\sqrt{29}$$

$\Rightarrow \left(\frac{210}{29} + \frac{10}{29}k\right)^2 + \left(\frac{4}{29}k + \frac{84}{29}\right)^2 = 464$

$\Rightarrow \left(\frac{1}{29}\right)^2 (210 + 10k)^2 + \left(\frac{1}{29}\right)^2 (4k + 84)^2 = 464$

$\Rightarrow (210 + 10k)^2 + (4k + 84)^2 = 390224$

$\Rightarrow 100k^2 + 4200k + 44100 + 16k^2 + 672k + 7056 = 390224$

$\Rightarrow 116k^2 + 4872k - 339068 = 0$

$\Rightarrow k^2 + 42k - 2923 = 0$

$\Rightarrow (k + 79)(k - 37) = 0$

$\Rightarrow k = -79$ or $k = 37$

Alternative Method (using Vectors):

Choose a point on l_1 so let A be the point with coordinates $(-4, k)$

Let P be a point on the line l_2, so $\overrightarrow{OP} = \begin{pmatrix} 8 + 2\mu \\ 9 + 5\mu \end{pmatrix}$

$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 8 + 2\mu \\ 9 + 5\mu \end{pmatrix} - \begin{pmatrix} -4 \\ k \end{pmatrix} = \begin{pmatrix} 12 + 2\mu \\ 9 - k + 5\mu \end{pmatrix}$

$\overrightarrow{AP} \cdot \begin{pmatrix} 2 \\ 5 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 12+2\mu \\ 9-k+5\mu \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 5 \end{pmatrix} = 0$

$\Rightarrow (12+2\mu) \times 2 + (9-k+5\mu) \times 5 = 0$

$\Rightarrow 24 + 4\mu + 45 - 5k + 25\mu = 0$

$\Rightarrow 29\mu + 69 - 5k = 0$

$\mu = \frac{5}{29}k - \frac{69}{29}$

So $\overrightarrow{AP} = \begin{pmatrix} 12 + 2\left(\frac{5}{29}k - \frac{69}{29}\right) \\ 9 - k + 5\left(\frac{5}{29}k - \frac{69}{29}\right) \end{pmatrix} = \begin{pmatrix} \frac{10}{29}k + \frac{210}{29} \\ -\frac{4}{29}k - \frac{84}{29} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = 4\sqrt{29} \Rightarrow \sqrt{\left(\frac{10}{29}k + \frac{210}{29}\right)^2 + \left(-\frac{4}{29}k - \frac{84}{29}\right)^2} = 4\sqrt{29}$

$\Rightarrow \left(\frac{1}{29}\right)^2 (10k+210)^2 + \left(\frac{1}{29}\right)^2 (4k+84)^2 = 464$

$\Rightarrow (10k+210)^2 + (4k+84)^2 = 390224$

$\Rightarrow 100k^2 + 4200k + 44100 + 16k^2 + 672k + 7056 = 390224$

$\Rightarrow 116k^2 + 4872k - 339068 = 0$

$\Rightarrow k^2 + 42k - 2923 = 0$

$\Rightarrow (k+79)(k-37) = 0$

$\Rightarrow k = -79$ or $k = 37$

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Convert to a 3D problem:

Let $\mathbf{p} = \begin{pmatrix} -4 \\ k \\ 0 \end{pmatrix}$

Let A be a known point on the line l_2 so that its position vector is $\mathbf{a} = \begin{pmatrix} 8 \\ 9 \\ 0 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} \Rightarrow \frac{\left|\begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix} \times \left(\begin{pmatrix} -4 \\ k \\ 0 \end{pmatrix} - \begin{pmatrix} 8 \\ 9 \\ 0 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix}\right|} = 4\sqrt{29} \Rightarrow \frac{\left|\begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix} \times \begin{pmatrix} -12 \\ k-9 \\ 0 \end{pmatrix}\right|}{\left|\begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix}\right|} = 4\sqrt{29}$

$\begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix} \times \begin{pmatrix} -12 \\ k-9 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 5 & 0 \\ -12 & k-9 & 0 \end{vmatrix}$

$= \big(5 \times 0 - 0 \times (k-9)\big)\mathbf{i} - \big(2 \times 0 - 0 \times (-12)\big)\mathbf{j} + \big(2 \times (k-9) - 5 \times (-12)\big)\mathbf{k}$

$= 0\mathbf{i} + 0\mathbf{j} + (2k + 42)\mathbf{k}$

$\therefore h = \frac{\left|\begin{pmatrix} 0 \\ 0 \\ 2k+42 \end{pmatrix}\right|}{\left|\begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix}\right|} = 4\sqrt{29} \Rightarrow \frac{\sqrt{0^2+0^2+(2k+42)^2}}{\sqrt{2^2+5^2+0^2}} = 4\sqrt{29} \Rightarrow \frac{\sqrt{(2k+42)^2}}{\sqrt{29}} = 4\sqrt{29}$

$\Rightarrow \sqrt{(2k+42)^2} = 116 \Rightarrow 2k + 42 = \pm 116$

$2k + 42 = 116 \Rightarrow k = 37$

Or $2k + 42 = -116 \Rightarrow k = -79$

So $k = -79$ or $k = 37$